Proposed Manuscript Insertions IV
Full Trace-Response Plumbing and Exact Equipotential Collapse

Research draft for the marked-width MLC program

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Abstract

A long planar plumbing neck is not, in general, a single scalar resistor: nonconstant boundary traces carry a nonvanishing half-cylinder response. We correct the finite-edge approximation by computing the complete Dirichlet-to-Neumann trace operator on the canonical ray/equipotential rectangle. Serial collars form an exact response semigroup, the full operator is determined by total resistance and a universal tangential operator, and its trace-renormalized nodal approximation has sharp relative error $2/(e^{\pi R}-1)$. The estimate survives arbitrary Hilbert-space Schur complements and preserves strict response contraction after an explicit finite depth choice. Exact scalar collapse is characterized: it holds precisely when all admissible traces are constant. These results close the direct-decoration end-plumbing trace-response gate but do not prove the global response theorem, MBTT, rigidity, or MLC.

1 Correction to the scalar plumbing model

Let

$$C_R = (0,R)\times(0,1)$$

with natural Neumann conditions on the side walls. Prescribe

$$f(y)=\sqrt{2}\cos(\pi y), \qquad g(y)=0$$

on the two end cross-sections. Both traces have mean zero, while their harmonic extension has energy

$$\mathcal{E}_R(f,0)=\pi\coth(\pi R)\longrightarrow\pi. \tag{1.1}$$

A model retaining only the two trace averages assigns zero energy. Thus a finite conductance edge represents the continuum neck only after the cross-sections have been electrically shorted, or after all admissible traces have separately been proved constant.

Warning 1.1 (Correction of the previous nodal estimate)**.** The node `ADAPTIVE_NODAL_RESPONSE_APPROXIMATION` from the preceding research insertion must be replaced. Its single-edge estimate is valid under an explicit equipotential-electrode hypothesis. For arbitrary surface traces, the full operator below is required.

2 Canonical trace collars

Let Φ be the Bottcher coordinate in the basin of infinity, and write

$$G(z)=\log|\Phi(z)|, \qquad \Theta(z)=\frac{1}{2\pi}\arg\Phi(z) \pmod 1.$$

A sector bounded by rays $\Theta=\theta_0,\theta_1$ and equipotentials $G=t_-,t_+$ is mapped to a width-one rectangle by

$$(x,y)=\left(\frac{G-t_-}{2\pi(\theta_1-\theta_0)},\frac{\Theta-\theta_0}{\theta_1-\theta_0}\right).$$

Its length is

$$R=\frac{t_+-t_-}{2\pi(\theta_1-\theta_0)}. \tag{2.1}$$

For a univalent branch of f^q,

$$G(f^q z) = 2^q G(z), \qquad \Theta(f^q z) = 2^q \Theta(z) \pmod 1. \tag{2.2}$$

Hence inverse branches scale potential thickness and angular width by the same factor, preserving the normalized trace coordinate.

Theorem 2.1 (Trace-compatible Bottcher collar)**.** *Choose a parent equipotential cross-section γ and put $\gamma_w = \phi_w(\gamma)$. Let $C_{w,a}$ be the full external sector between γ_w and γ_{wa}, with the two authorized ray sides retained as walls. Then $C_{w,a}$ is marked-conformally a rectangle of width one and some length $R_a > 0$. Adjacent child and parent trace parameterizations agree. Thus a tail $a_1 \cdots a_m$ is marked-conformally a rectangle of length*

$$R(a_1 \cdots a_m) = \sum_{j=1}^{m} R_{a_j}. \tag{2.3}$$

The construction is natural under every certified univalent presentation and root enlargement that retains the collar.

Remark 2.2 (Source-facing check). A claimed application must verify that no protected collar, obstacle, or localization boundary cuts through the selected external sector. Failure is recorded at the first affected cell.

3 Exact full trace response

Let

$$\varphi_0(y) = 1, \qquad \varphi_n(y) = \sqrt{2}\cos(n\pi y), \quad n \geq 1,$$

be the Neumann eigenbasis on $(0,1)$. Put

$$A = \sqrt{-\partial_{yN}^2}, \qquad A\varphi_0 = 0, \quad A\varphi_n = n\pi\varphi_n, \tag{3.1}$$

and let P_0 be projection onto constants. The normalized trace space is $\mathcal{H} = H_N^{1/2}(0,1)$ with

$$\|f\|_{\mathcal{H}}^2 = |f_0|^2 + \sum_{n\geq 1} n\pi |f_n|^2. \tag{3.2}$$

Theorem 3.1 (Full rectangle Dirichlet-to-Neumann formula)**.** *For end traces $f, g \in \mathcal{H}$, the minimizing energy on C_R is*

$$\begin{aligned}\mathcal{E}_R(f,g) = {} & \frac{|f_0 - g_0|^2}{R} \\ & + \sum_{n\geq 1} n\pi \left[\coth(n\pi R)(|f_n|^2 + |g_n|^2) - 2\,\mathrm{csch}(n\pi R)\Re(f_n\overline{g_n})\right].\end{aligned} \tag{3.3}$$

Equivalently,

$$\mathcal{N}_R = \begin{pmatrix} \alpha_R(A) & -\beta_R(A) \\ -\beta_R(A) & \alpha_R(A) \end{pmatrix}, \tag{3.4}$$

where

$$\alpha_R(\lambda) = \lambda\coth(R\lambda), \qquad \beta_R(\lambda) = \lambda\,\mathrm{csch}(R\lambda), \tag{3.5}$$

and both functions take the value R^{-1} at $\lambda = 0$.

Proof. For $n = 0$, the minimizing mode is linear in x. For $n \geq 1$, it is

$$u_n(x) = \frac{\sinh(n\pi(R-x))}{\sinh(n\pi R)} f_n + \frac{\sinh(n\pi x)}{\sinh(n\pi R)} g_n.$$

Green's identity gives the displayed mode energy. Orthogonality completes the sum. □

Proposition 3.2 (Marked phase)**.** *If the right trace is identified with the left normalized coordinate by a unitary U commuting with A and preserving constants, then*

$$\mathcal{N}_{R,U} = \begin{pmatrix} \alpha_R(A) & -\beta_R(A)U \\ -U^*\beta_R(A) & U^*\alpha_R(A)U \end{pmatrix}. \tag{3.6}$$

Serial gluing multiplies the phases. For R bounded below, the family over a compact marked phase group is norm compact; the phase-dependent off-diagonal part is smoothing.

Theorem 3.3 (Exact response semigroup)**.** *If two trace-compatible collars of lengths R, S are glued along their common parameterized cross-section, then*

$$\boxed{\mathcal{N}_R \star \mathcal{N}_S = \mathcal{N}_{R+S},} \tag{3.7}$$

where $\star$ means addition of energy forms followed by minimization over the common trace.

Proof. The glued surface is the rectangle of length $R+S$. Restricting its harmonic minimizer gives the minimizing interface trace, and conversely the two minimizers glue to an admissible function on the long rectangle. □

Corollary 3.4 (Finite encoding of the full operator)**.** *The complete trace response of a serial tail is determined exactly by its total resistance R, the fixed tangential operator A, and the marked phase/orientation. It is not an arbitrary infinite-dimensional state.*

4 Exact equipotential collapse

Restricting (3.3) to constant traces gives

$$\mathcal{E}_R(a,b) = \frac{|a-b|^2}{R}. \tag{4.1}$$

Thus the constant mode is exactly a scalar resistor of conductance $1/R$.

Theorem 4.1 (Exact mean-collapse criterion)**.** *Let $V \subset \mathcal{H}$ be a linear admissible trace subspace and let $M(f) = f_0$ be the trace-mean map. The restriction of $\mathcal{E}_R$ to $V \oplus V$ factors through the two means if and only if $M|_V$ is injective. If constants are admissible, $\mathbb{C}\varphi_0 \subset V$, or more generally if $f - M(f)\varphi_0 \in V$ for every $f \in V$, this is equivalent to*

$$V = \mathbb{C}\varphi_0 = \ker A. \tag{4.2}$$

Proof. If the response factors through means and $M(f) = 0$, then $\mathcal{E}_R(f,0) = \mathcal{E}_R(0,0) = 0$. Formula (3.3) forces $f = 0$, so $M|_V$ is injective. Conversely, injectivity gives a unique representative $f_a \in V$ of each $a \in M(V)$; define $q_R(a,b) = \mathcal{E}_R(f_a, f_b)$. If constants are admissible, then $f - M(f)\varphi_0 \in V \cap \ker M$, hence it vanishes. The same argument uses only mean-subtraction invariance. □

Remark 4.2 (One-dimensional shape exception). Without constant admissibility or mean-subtraction invariance, a nonconstant one-dimensional trace space can have injective mean map and therefore admit a one-scalar encoding. This records one fixed nonconstant shape; it is not an electrically shorted equipotential.

Corollary 4.3 (Shorted cross-section)**.** *If a cross-section is declared one perfect electrical electrode, every admissible trace is constant and the one-edge model is exact.*

Warning 4.4. A geometric Bottcher equipotential is not automatically an electrical equipotential for the response problem.

5 Trace-renormalized nodal response

The mean-zero response of a semi-infinite rectangle is A. Therefore define

$$\mathcal{E}_R^{\#}(f,g) = \frac{|f_0 - g_0|^2}{R} + \langle Af, f\rangle + \langle Ag, g\rangle, \tag{5.1}$$

with operator

$$\mathcal{N}_R^{\#} = \begin{pmatrix} A + R^{-1}P_0 & -R^{-1}P_0 \\ -R^{-1}P_0 & A + R^{-1}P_0 \end{pmatrix}. \tag{5.2}$$

This is the correct nodal normalization: two semi-infinite trace ends plus the exact resistor joining their constant modes.

Theorem 5.1 (Sharp full trace estimate)**.** *Put*

$$\eta(R) = \frac{2}{e^{\pi R} - 1}. \tag{5.3}$$

Then

$$\left| \mathcal{E}_R - \mathcal{E}_R^{\#} \right| \le \eta(R) \left(\langle Af, f \rangle + \langle Ag, g \rangle \right). \tag{5.4}$$

Whenever $\eta(R) < 1$,

$$\boxed{(1 - \eta(R))\mathcal{N}_R^{\#} \preceq \mathcal{N}_R \preceq (1 + \eta(R))\mathcal{N}_R^{\#}.} \tag{5.5}$$

The modewise constant is sharp.

Proof. For $t = n\pi R$, the normalized difference matrix is

$$\begin{pmatrix} \coth t - 1 & -\operatorname{csch} t \\ -\operatorname{csch} t & \coth t - 1 \end{pmatrix}.$$

Its eigenvalues are

$$\frac{2}{e^t - 1}, \qquad -\frac{2}{e^t + 1}.$$

The largest absolute value is maximized at $n = 1$. The constant modes agree exactly. □

Corollary 5.2 (Explicit depth)**.** *To obtain relative trace error at most* $\delta \in (0, 1)$*, it suffices that*

$$R \ge \frac{1}{\pi} \log \left(1 + \frac{2}{\delta} \right). \tag{5.6}$$

If every chosen cell has resistance at least $R_* > 0$*, it suffices to take*

$$\ell \ge \left\lceil \frac{1}{\pi R_*} \log \left(1 + \frac{2}{\delta} \right) \right\rceil. \tag{5.7}$$

6 Compact operator limit

Let

$$\mathcal{N}_\infty = \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix}, \tag{6.1}$$

with A vanishing on constants.

Theorem 6.1 (Full trace compactification)**.** *In the normalized* $\mathcal{H} \oplus \mathcal{H}$ *to dual operator norm,*

$$\| \mathcal{N}_R - \mathcal{N}_\infty \|_{\mathrm{op}} \le \max \left\{ \frac{2}{R}, \frac{2}{e^{\pi R} - 1} \right\}. \tag{6.2}$$

Hence $R \mapsto \mathcal{N}_R$ *extends continuously to* $R = \infty$*.*

Proof. The constant block has eigenvalues $0, 2/R$. The mean-zero block is controlled by the preceding theorem. □

Remark 6.2. The end tail therefore contributes a compact one-parameter family of complete trace operators. The root-side joint trace response is transported exactly; no new arbitrary operator complexity is created by the address depth.

7 Hilbert-space gluing and external response

Let $Q(x,\phi) \geq 0$ be the component energy for external terminal values x and interface trace ϕ. For a neck form N, define

$$q_N(x) = \inf_{\phi}\{Q(x,\phi) + N(\phi)\}. \tag{7.1}$$

Lemma 7.1 (Relative form bounds survive elimination)**.** *If*

$$(1-\delta)N_0 \preceq N_1 \preceq (1+\delta)N_0, \qquad 0 \leq \delta < 1,$$

then

$$(1-\delta)q_{N_0} \leq q_{N_1} \leq (1+\delta)q_{N_0}. \tag{7.2}$$

Proof. Add the same nonnegative form Q and take the infimum over the same trace space. □

For disjoint tails $R_1, \ldots, R_s$, put

$$\eta_* = \max_i \frac{2}{e^{\pi R_i} - 1}. \tag{7.3}$$

Theorem 7.2 (Physical versus trace-renormalized response)**.** *Let Λ_{phys} be the external response obtained with the exact full necks, and let $\Lambda_{\#}$ be obtained with the trace-renormalized neck forms. On the quotient by global constants,*

$$\boxed{(1-\eta_*)\Lambda_{\#} \preceq \Lambda_{\mathrm{phys}} \preceq (1+\eta_*)\Lambda_{\#}.} \tag{7.4}$$

The same estimate holds after every fixed finite-electrode compression.

Proof. Sum the neck estimates, add the component energies, and apply the lemma. Compression by an electrode map preserves Loewner order. □

8 Preservation of a strict transition

Let L be a nonnegative, positively homogeneous, Loewner-monotone gauge. Then (7.4) implies

$$(1-\eta_*)L(\Lambda_{\#}) \leq L(\Lambda_{\mathrm{phys}}) \leq (1+\eta_*)L(\Lambda_{\#}). \tag{8.1}$$

Theorem 8.1 (Strict response transition survives)**.** *Suppose the trace-renormalized normalized block satisfies*

$$L_{\#}^{\mathrm{out}} \leq \mu L_{\#}^{\mathrm{in}} + E, \qquad 0 < \mu < 1. \tag{8.2}$$

If both endpoint materializations have trace error at most η, then

$$L_{\mathrm{phys}}^{\mathrm{out}} \leq \mu\frac{1+\eta}{1-\eta} L_{\mathrm{phys}}^{\mathrm{in}} + (1+\eta)E. \tag{8.3}$$

The homogeneous factor remains strictly below one whenever

$$\eta < \frac{1-\mu}{1+\mu}. \tag{8.4}$$

Equivalently, a backward factor $\lambda = \mu^{-1} > 1$ becomes

$$\lambda_{\mathrm{eff}} = \lambda\frac{1-\eta}{1+\eta} > 1. \tag{8.5}$$

Remark 8.2 (Exact retention). If the state retains $\mathcal{N}_R$ through the exact functional calculus (3.4), the plumbing approximation error is zero. The relative estimate is needed only when passing to the compact trace-renormalized nodal model.

9 Promotion statement

Theorem 9.1 (End plumbing trace-response control)**.** *For every source-authorized direct-decoration tail whose ray/equipotential collar avoids all protected data and remains in the localization:*

1. *the complete full trace response is exactly $\mathcal{N}_R$;*
2. *serial gluing is exact and depends only on total resistance and marked phase;*
3. *the operator family is compact at infinite depth;*
4. *the trace-renormalized response has the sharp relative error (5.3);*
5. *arbitrary component elimination and finite-electrode compression retain the same comparison;*
6. *strict response contraction survives after the explicit depth choice; and*
7. *scalar collapse is invoked only under the exact constant-trace criterion.*

Thus `END_PLUMBING_TRACE_RESPONSE_CONTROL` *is proved in the direct-decoration trace-compatible category.*

Remark 9.2 (Remaining global gate). The theorem does not supply a trace collar for every mixed virtual-Molecule block, a universally bounded response/WAD/Value quotient, a common KKL-neutral- virtual chart, MBTT, coercivity, rigidity, or MLC. The broad node `H2_RESPONSE_REALIZATION` remains open.

10 Ledger correction and next theorem

The previous single-edge node should be retained only as `SHORTED_EQUIPOTENTIAL_NODAL_RESPONSE`. New proved nodes are:

1. `BOTTCHER_TRACE_COLLAR_ATLAS`;
2. `FULL_TRACE_RECTANGLE_DTN`;
3. `TRACE_RESPONSE_SEMIGROUP`;
4. `EQUIPOTENTIAL_COLLAPSE_CRITERION`;
5. `TRACE_RENORMALIZED_NECK_ESTIMATE`;
6. `END_TRACE_RESPONSE_COMPACTIFICATION`;
7. `HILBERT_TRACE_GLUE_CALCULUS`;
8. `TRACE_STABLE_STRICT_TRANSITION_TRANSFER`; and
9. `END_PLUMBING_TRACE_RESPONSE_CONTROL`.

The next source-facing target is a universal trace-interface authorization and bounded future response theorem: every mixed block must supply its protected trace collars, exact component joint response forms, enlargement maps, a future-complete terminal/Value quotient, and one calibrated chart in which the remaining non-plumbing error stays below the strict transition margin.